

Juan Esteban Beron Zapata

Incoming M.S. Student in Aerospace Engineering

GitHub | LinkedIn | Portfolio / Webpage

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Personal Profile

Aeronautical Engineering graduate with three years of research and development experience in autonomous aerospace systems. My work centers on safety-critical and resilient autonomy for aerospace vehicles operating in degraded, dynamic, and uncertain environments. I have worked across nonlinear and optimization-based control, control barrier and Lyapunov methods, perception, multi-robot systems, communication-aware autonomy, and spacecrafts simulation and hardware testing. My long-term interest is to develop autonomous aerospace systems that can explore challenging environments while maintaining safety and mission capability under sensing degradation, communication constraints, faults, and changing operating conditions.

Education

The University of Texas at Austin

Spring 2027–

M.S. in Aerospace Engineering, Incoming Student

- Research interests: autonomous systems, safety-critical control, resilient aerospace systems, and robotics.

Universidad Pontificia Bolivariana (UPB), Colombia

2021–2026

B.Eng. in Aerospace Engineering

- Cumulative GPA: 3.84/5.00.
- Relevant coursework: Flight Dynamics and Control, Aerial Robotics, Avionics, Computational Fluid Dynamics, Numerical Methods, Aircraft Design, Aerodynamics, and Rocket & Satellite Engineering.

Research Experience

California Institute of Technology (Caltech), California — AMBER Lab

Jun 2026–Sep 2026

SURF Research Fellow — Supervised by Yana Lishkova, Ph.D., and Prof. Aaron D. Ames

- Contributed to ATMOS, a free-flying spacecraft testbed, developing ROS 2/Gazebo/PX4 simulation and hardware-validation workflows with motion tracking and IoT instrumentation.
- Developed and evaluated a safety-critical control architecture for a Mars-helicopter landing analog in dynamic obstacle environments using HOCBFs, control Lyapunov functions, Poisson/PDE-based safety fields, and contingency-aware control logic.
- Implemented LQR-tree “funnels” as contingency-aware safety filters, generating safe trajectories in 5–10 ms compared with 20–70 ms for Poisson- and Hamilton–Jacobi-based methods in our test scenarios; validated the approach in simulation and hardware experiments, while formal theoretical validation remains ongoing.

King Abdullah University of Science and Technology (KAUST), Saudi Arabia — NetLab

Apr 2026–Sep 2026

VSRR Research Fellow — Supervised by Prof. Basem Shihada and Dr. Balsam Alkouz

- Developed a drone-swarm networking and autonomy framework integrating ROS 2, PX4, Isaac Sim, and Sionna for communication-aware multi-robot experiments.
- Studied resilient swarm operation under communication degradation and UAV failures, including dynamic topology adaptation, standby integration, and recovery of end-to-end connectivity.
- Performed analogous simulations and comparative validation against AirPAW datasets, evaluating SNR, capacity, routing behavior, and network performance

National Central University (NCU), Taiwan — SenDiMoniProg Lab

Sep 2025–Jan 2026

TEEP & IIPP Research Intern — Supervised by Prof. Min-Chun Pan

- Built a ROS 2 multi-sensor fusion stack using IMU, GNSS, camera-VIO, and odometry on Jetson-class hardware; implemented EKF/UKF estimation with SLAM/VIO fallback.
- Validated navigation performance at the ARTC proving ground and in laboratory track experiments under degraded sensing conditions.
- Optimized the ESP32–Jetson–server telemetry pipeline, reducing end-to-end latency from 92 ms to 6 ms and improving communication reliability in field tests.

Universidad de Antioquia (UdeA), Colombia — CubeSat-Class Atmospheric Probe

Apr 2024–Jul 2025

Research Contributor in Flight Dynamics — Supervised by Prof. Francisco Hidalgo

- Modeled atmospheric descent dynamics, including drag, attitude, and stability, and designed a multi-Pitot airspeed-vector sensing system.
- Fused IMU, GPS, and Pitot measurements for post-flight trajectory reconstruction and supported airflow-correction modeling, CFD analysis, and wind-tunnel calibration.
- Supported avionics, telemetry, and launch-day operations for the Spaceport America Cup 2025, validating aerodynamic and flight-dynamics models with real flight data.

Work Experience

Daetech (Medellín, Colombia)

Jan 2023–Sep 2024

UAV Simulation Developer in Autonomy

- Developed a ROS 2-based UAV simulation framework with high-fidelity dynamics, real-time (60 Hz) execution, and photorealistic urban environments for testing navigation and autonomy algorithms.

Selected Technical Projects

ATMOS — Autonomy Testbed for Multi-purpose Orbiting Systems 2026

- Contributed to the ATMOS free-flying spacecraft testbed through repository integration, ROS 2/Gazebo/PX4 simulation, motion-capture interfaces, IoT instrumentation, and hardware-validation workflows.

SWARMSYM — Drone Swarm Networking & SNaaS Framework 2026

- Developed a ROS 2/PX4/Isaac Sim/Sionna framework for reproducible multi-UAV and SNaaS simulations, evaluating network topology, SNR, capacity, routing, communication-aware autonomy, UAV failures, and recovery strategies. [GitHub]

VOLTA Rocketry — Spaceport America Cup, 10k COTS 2024–2025

- Integrated payload, sensing, and telemetry systems for the Spaceport America Cup 2025, enabling flight-data acquisition and post-flight trajectory reconstruction for aerodynamic and performance analysis.

AIAA Design/Build/Fly — UPB 2021–2022

- Designed propulsion matching and aircraft electrical architecture for the DBF competition aircraft; qualified and flew at the Wichita fly-off, finishing 14th of 110 teams worldwide.

Academic Aerospace Design Projects 2021–2024

- Completed projects in canard-aircraft design, annular-nozzle combustion testing, Pilatus PC-12 electrical-system modeling, conceptual electronics for an autonomous electric agricultural aircraft, and ANSYS CFD analysis of high-speed airfoils.

Awards, Honors & Funding

- Caltech SURF Award — Summer Undergraduate Research Fellowship, 2026; \$8,410 research award.
- KAUST VSRP Fellowship — Visiting Student Research Program, 2026; award code #XXXXXXX; \$6,000.
- TEEP & IIPP Funding — National Central University, Taiwan, 2025; \$5,000 research support.
- Renault Foundation Scholarship — merit-based undergraduate scholarship, 2025; \$2,000.
- Generación E Scholarship — awarded for outstanding national high-school examination performance; full-tuition undergraduate scholarship with an estimated present value of approximately \$35,000.
- Young Researcher Recognition — Risaralda Department, 2023.

Skills

- Control & Autonomy: LQR, MPC, nonlinear control, CBFs/HOCBFs, CLFs, safety-critical control, contingency-aware planning, multi-robot autonomy.
- Robotics & State Estimation: ROS 2, PX4, VIO/SLAM, EKF/UKF sensor fusion, OptiTrack motion capture, Crazyflie platforms, UAV navigation.
- Simulation & Hardware Validation: Gazebo, Isaac Sim, Webots, HIL, mission replay, simulation-to-hardware validation, experimental testing.
- Wireless & Networked Systems: NVIDIA Sionna, multi-UAV networking, communication-aware autonomy, SNR/capacity analysis, routing, telemetry, IoT systems.
- Scientific Computing & Optimization: Python, C++, CasADi, NumPy, SciPy, Pandas, Matplotlib, MATLAB/Simulink, numerical optimization, signal processing.
- Embedded Systems & Instrumentation: Jetson, Raspberry Pi, ESP32, IMU, GNSS/GPS, barometric and Pitot sensing, data acquisition, hardware integration.
- Aerodynamics & Structural Analysis: ANSYS, OpenFOAM, CFD, basic FEA, aerodynamic modeling, wind-tunnel testing, sensor-placement studies.
- Software & Research Tools: Linux, Docker, Git, LaTeX/Overleaf, reproducible workflows, technical documentation.

Certifications & Standardized Exams

- Saber Pro (ICFES): 185; above the 94th percentile. Certificate/Exam ID: #XXXXXXX
- GRE General Test: Quantitative 167/170; Verbal 159/170. Exam ID: #XXXXXXX
- IELTS Academic: Reading 7.5; Listening 7.5; Speaking 7.5; Writing 6.0. Certificate ID: #XXXXXXX
- ICFES Saber 11: 385/500; top 2% nationally. Exam ID: #XXXXXXX